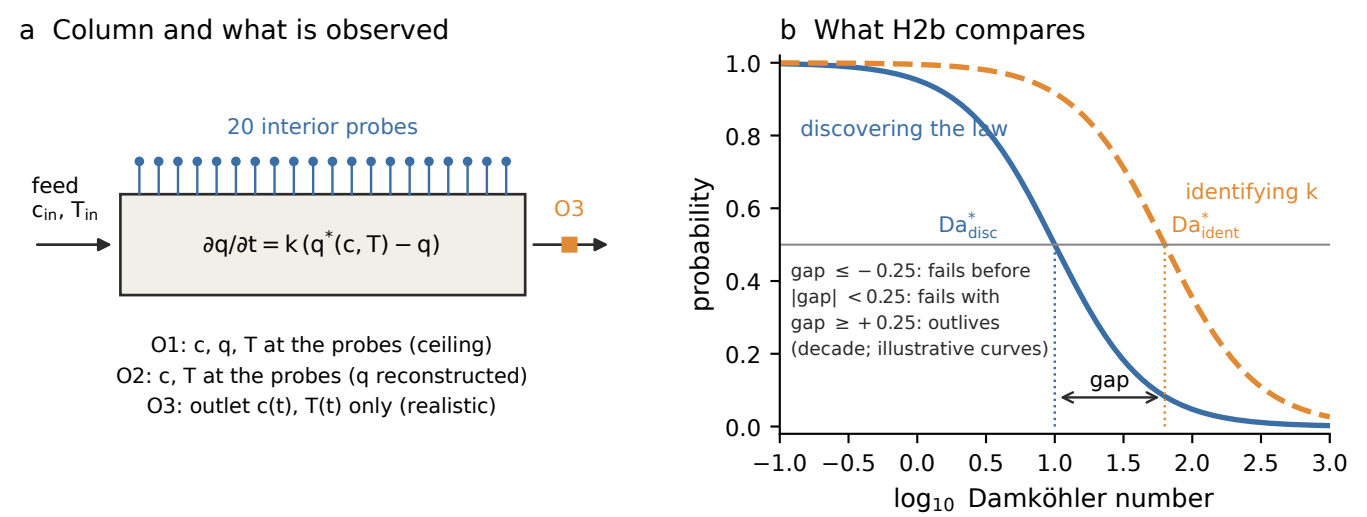
a Column and what is observed
20 interior probes
feed
c_{in}, T_{in}
$\partial q/\partial t = k\,(q^*(c,T) - q)$
O3
O1: c, q, T at the probes (ceiling)
O2: c, T at the probes (q reconstructed)
O3: outlet c(t), T(t) only (realistic)
b What H2b compares
probability
discovering the law
identifying k
Da^*_{disc}
Da^*_{ident}
gap ≤ -0.25: fails before
$|gap| < 0.25$: fails with
gap $\geq +0.25$: outlives
(decade; illustrative curves)
gap
1.0
0.8
0.6
0.4
0.2
0.0
−1.0 −0.5 0.0 0.5 1.0 1.5 2.0 2.5 3.0
$\log_{10}$ Damköhler number